Vidyamandir Classes
Gurukul for IITJEE & Medical Preparation

illuminati - 2019

Advanced Mathematics Assignment-1A

Calculus

Section 1	Single Correct Answer Type

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. The value of $\lim\limits_{x \to 0^+} \dfrac{e^{\left(x^x - 1\right)} - x^x}{\left[\left(x^2\right)^x - 1\right]^2}$ is equal to :

(A) 1 **(B)** 1/8 **(C)** 3/2 **(D)** 1/4

2. If $f(x) = max\left\{sin\, x,\, sin^{-1}(cos\, x)\right\}$, then :

(A) f is differentiable everywhere

(B) f is continuous every where but not differentiable

(C) f is discontinuous at $x = \dfrac{n\pi}{2}, n \in I$ **(D)** f is non-differentiable at $x = \dfrac{n\pi}{2}, n \in I$

3. Which of the following statement is true ?

(A) The equation $sin\, x - x = 0$ has a real root in $\left(\dfrac{\pi}{4}, \dfrac{\pi}{2}\right)$

(B) The equation $tan\, x - x = 0$ has a real root in $\left(\dfrac{\pi}{6}, \dfrac{\pi}{3}\right)$

(C) If $f(x)$ is a real-valued continuous function in [0, 2] then there exist some $c \in R$ such that $f(x) \ge (c)$ for all $x \in [0, 2]$

(D) If $g(x)$ is a real-valued function defined on [3, 5] and $g(3) \,.\, g(5) < 0$ then there exist some $\alpha \in (3, 5)$ such that $g(\alpha) = 0$

4. Let $f: R \to (0, \infty)$ be such that $f(x) + \dfrac{e^{x+x^2}}{f(x)} \le e^x + e^{x^2}\ \forall\ x > 0$, then $\lim\limits_{x \to 1} f(x)$ is :

(A) 1 **(B)** $1/e$ **(C)** e **(D)** $2e$

5. If $\Delta_r = \begin{vmatrix} r(r-1) & 2(r+2) & 2015 \\ r & 1 & -6 \\ r^2 & 4r & 0 \end{vmatrix}$ then $\lim\limits_{n \to \infty} \dfrac{\sum\limits_{r=1}^{n} \Delta_r}{n^4}$ is equal to :

(A) 2 **(B)** 3 **(C)** 4 **(D)** 5

6. The value of $\lim\limits_{x \to 0^+} \left(\dfrac{cot^{-1}\left(ln\{x\}\right)}{\pi}\right)^{-\ell n\{x\}}$ is equal to : [Note : where {.} denotes fractional part function]

(A) $e^{1/\pi}$ **(B)** $e^{-1/\pi}$ **(C)** $e^{-2/\pi}$ **(D)** $e^{2/\pi}$

7. Let $f(t)=|t|+|t-1|\ \forall\ t\in R$ and $g(x)=\begin{cases} max(f(t)), & x-1\le t\le x \quad 0\le x\le 1 \\ 3-x, & 1<x\le 2 \end{cases}$

Number of points where $g(x)$ is non-differentiable in [0, 2], is :

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

8. Let $\frac{\pi}{3}<A<\frac{\pi}{2}$ and a function $f(x)$ is defined as :

$$f(x)=\begin{cases} \lim\limits_{n\to\infty}\dfrac{ax^2\left(\sin A-\sin^3 A\right)-(5x-b)\left|\sin A-\sin^3 A\right|^n}{\left(\sin A-\sin^3 A\right)-\left|\sin A-\sin^3 A\right|^n}, & x\in Q \\ \lim\limits_{n\to\infty}\dfrac{ax^2\left(\sin A+\sin^3 A\right)+(5x-b)\left|\sin A+\sin^3 A\right|^n}{\left(\sin A+\sin^3 A\right)+\left|\sin A+\sin^3 A\right|^n}, & x\notin Q \end{cases}$$

If $f(x)$ is continuous at $x=2$ and $x=3$ then the value of $b-a$, is :

(A) 3 **(B)** 4 **(C)** 5 **(D)** 6

9. Let f be a differentiable function on $(0,\infty)$ and suppose that $\lim\limits_{x\to\infty}\left(f(x)+f'(x)\right)=L$ where L is a finite quantity, then which of the following must be true ?

(A) $\lim\limits_{x\to\infty} f(x)=0$ and $\lim\limits_{x\to\infty} f'(x)=L$ **(B)** $\lim\limits_{x\to\infty} f(x)=L/2$ and $\lim\limits_{x\to\infty} f'(x)=L/2$

(C) $\lim\limits_{x\to\infty} f(x)=L$ and $\lim\limits_{x\to\infty} f'(x)=0$ **(D)** Nothing definite can be said

10. Let f and g be twice differentiable function on R and $f''(5)=8,\ g''(5)=2$ then $\lim\limits_{x\to 5}\left[\dfrac{f(x)-f(5)-(x-5)f'(5)}{g(x)-g(5)-(x-5)g'(5)}\right]$ is equal to :

(A) 0 **(B)** 1 **(C)** 2 **(D)** 4

11. Let f be derivable function $\forall\ x\in R$ such that $f\left(\dfrac{x+y}{2}\right)=\dfrac{f(x)+f(y)}{2};\ \forall\ x,\ y\in R$. If $f'(0)=-1$ and $f(0)=1$, then :

(A) $2f^{-1}(x)=f(x)$ **(B)** $f^{-1}(x)=2f(x)$

(C) $f^{-1}(x)=-f(x)$ ~~**(D)**~~ $f^{-1}(x)=f(x)$

12. If $h(x)$ is inverse of $g(x)$ and $g'(x)=\dfrac{1}{1+x^3}$ then $h''(x)$ is equal to :

(A) $\dfrac{-3[h(x)]^2}{\left[1+\{h(x)\}^3\right]^3}$ **(B)** $\dfrac{-3[h(x)]^2}{\left[1+\{h(x)\}^3\right]^2}$ **(C)** $\dfrac{-3[h(x)]^2}{\left[1+\{h(x)\}^3\right]}$ **(D)** $3h^2(x)\left[1+h^3(x)\right]$

13. Let $f(x)$ be a continuous and period function such that $f(x)=f(x+T)$ for all $x\in R, T>0$. If $\int_{-2T}^{a+5T} f(x)dx=19\ (a>0)$ and $\int_0^T f(x)dx=2$, then $\int_0^a f(x)dx$ is equal to :

(A) 3 **(B)** 5 **(C)** 7 **(D)** 9

14. Let $\alpha>-1$ and $\beta>-1$, then the value of $\lim\limits_{n\to\infty} n^{\beta-\alpha}\left(\dfrac{1^\alpha+2^\alpha+....+n^\alpha}{1^\beta+2^\beta+....+n^\beta}\right)$ is :

~~**(A)**~~ $\dfrac{\beta+1}{\alpha+1}$ **(B)** $\dfrac{\alpha+1}{\beta+1}$ **(C)** $\dfrac{\alpha+2}{\beta+2}$ **(D)** $\dfrac{\beta+2}{\alpha+2}$

15. If x and y are independent variables and $f(x)=\left(\int_0^x e^{x-y} f'(x)dy\right)-\left(x^2-x+1\right)e^x$, where $f(x)$ is a differentiable function, then $f\left(\frac{-1}{2}\right)$ equals :

(A) $2\sqrt{e}$ **(B)** $-2\sqrt{e}$ **(C)** $-2/\sqrt{e}$ **(D)** $2/\sqrt{e}$

16. Let P_k be a point in xy plane whose x coordinate is $1+\frac{k}{n}(k=1, 2, 3, \ldots, n)$ on the curve $y=\ell nx$. If A is (1, 0), then $\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{k=1}^{n}\left(AP_k\right)^2$ equals :

(A) $\frac{1}{3}+2\ell n^2 2$ **(B)** $\frac{1}{3}+2\ell n^2\left(\frac{2}{e}\right)$ **(C)** $\frac{1}{3}+\ell n^2\left(\frac{2}{e}\right)$ **(D)** $\frac{1}{3}+2\ell n\left(\frac{2}{e}\right)$

17. $\int_0^1\left(\sqrt[4]{1-x^7}-\sqrt[7]{1-x^4}\right)dx$ is equal to :

(A) $1/2$ **(B)** 1 **(C)** 0 **(D)** None of these

18. The value of $\lim\limits_{n\to\infty}\left\{\frac{1}{n^4}\left(\sum\limits_{k=1}^{n}k^2\int_k^{k+1}x\,\ell n\left[(x-k)(k+1-x)\right]dx\right)\right\}$ is :

(A) 1 **(B)** $\frac{1}{2}$ **(C)** $\frac{-1}{2}$ **(D)** -1

19. Given a curve C. Suppose that the tangent line at P(x, y) on C is perpendicular to the line joining P and Q(1, 0). If the line $2x+3y-15=0$ is tangent to the curve C then the curve C denotes :

(A) a circle touching the x-axis
(B) a circle touching the y-axis
(C) circle whose y-intercept a $4\sqrt{3}$
(D) a parabola with axis parabola to y-axis

20. A function $y=f(x)$ satisfies the condition $f'(x)\sin x+f(x)\cos x=1$, $f(x)$ being bounded when $x\to 0$. If $I=\int_0^{\pi/2}f(x)dx$ then :

(A) $\frac{\pi}{2}<I<\frac{\pi^2}{4}$ **(B)** $\frac{\pi}{4}<I<\frac{\pi^2}{2}$ **(C)** $1<I<\frac{\pi}{2}$ **(D)** $0<I<1$

21. Let g be a differentiable function satisfying $\int_0^x (x-t+1)g(t)dt=x^4+x^2$ for all $x\geq 0$. The value of $\int_0^1\frac{12}{g'(x)+g(x)+10}dx$ is equal to :

(A) $\pi/6$ **(B)** $\pi/3$ **(C)** $\pi/4$ **(D)** $\pi/2$

22. Let a solution $y=y(x)$ of the differential equation, $dy+xy\,dx=x$ satisfy $y(0)=2$, then the area enclosed by the curve $y=y(x)$, $x=0$ and $y=1$ in the first quadrant, is :

(A) $\int_0^\infty e^{-\frac{x^2}{2}}dx$ **(B)** $2\int_0^\infty e^{-\frac{x^2}{2}}dx$ **(C)** $\int_0^\infty\left(1+e^{-\frac{x^2}{2}}\right)dx$ **(D)** $2\int_0^\infty\left(1+e^{-\frac{x^2}{2}}\right)dx$

23. Let $y'(x)+\frac{g'(x)}{g(x)}.\,y(x)=\frac{g'(x)}{1+g^2(x)}$ where $f'(x)$ denotes $\frac{df(x)}{dx}$ and $g(x)$ is a given non-constant differentiable function on R. If $g(1)=y(1)=1$ and $g(e)=\sqrt{(2e-1)}$ then $y(e)$ equals :

(A) $\frac{3}{2g(e)}$ **(B)** $\frac{1}{2g(e)}$ **(C)** $\frac{2}{3g(e)}$ **(D)** $\frac{1}{3g(e)}$

24. A continuous function $f : R \to R$ satisfy the differential equation $f(x) = (1+x^2)\left[1 + \int_0^x \frac{f^2(t)}{1+t^2}dt\right]$ then the value of $f(-2)$ is :

(A) 0 **(B)** 17/15 **(C)** –17/15 **(D)** 15/17

25. Let $f : R^+ \to R$ be a differentiable function satisfying $f(x) = e + (1-x)ln\left(\frac{x}{e}\right) + \int_1^x f(t)dt$ for all $x \in R^+$, then number of points of local extrema of $f(x)$ is :

(A) f has maxima but no minima
(B) f has minima but no maxima
(C) f has neither maxima nor minima
(D) f has both maxima and minima

26. Let Z be a complex number such that $Re(Z) = \sqrt{x^2+4}$ and $Im(Z) = \sqrt{y-4}$ satisfying $|Z| = \sqrt{10}$. Area enclosed by the set of points (x, y) on the complex pane, is :

(A) $8\sqrt{6}$ **(B)** $4\sqrt{6}$ **(C)** $20\sqrt{10}/3$ **(D)** $40\sqrt{10}/3$

27. The graph of $f(x) = x^2$ and $g(x) = cx^3\ (c > 0)$ intersect at the points (0, 0) and $\left(\frac{1}{c}, \frac{1}{c^2}\right)$. If the region which lies between these graphs and over the interval $\left(0, \frac{1}{c}\right)$ has the area equal to $\frac{2}{3}$ then the value of c is :

(A) 1 **(B)** 1/3 **(C)** 1/2 **(D)** 2

28. Consider the following regions in the plane: $R_1 \equiv \{(x, y): 0 \le x \le 1 \text{ and } 0 \le y \le 1\}$; $R_2 \equiv \left\{(x, y): x^2 + y^2 \le \frac{4}{3}\right\}$

The area of the region $R_1 \cap R_2$ can be expressed as $\frac{a\sqrt{3} + b\pi}{9}$, where a and b are integers. Then the value of $(a + b)$ equals :

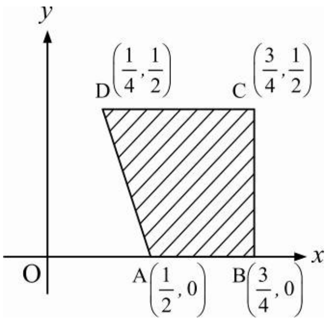
(A) 2 **(B)** 3 **(C)** 4 **(D)** 6

29. The slope of the tangent to the curve $y = f(x)$ at $[x, f(x)]$ is $(2x + 1)$. If the curve passes through the point (1, 3) then the area bounded by the curve $y = f(x)$ and the normal to the curve $y = (1+x)^y + sin^{-1}(sin^2 x)$ at $x = 0$, is equal to :

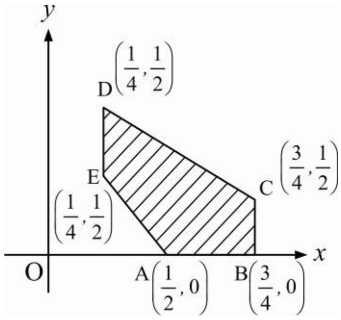
(A) 5/6 **(B)** 6/5 **(C)** 3/4 **(D)** 4/3

30. Which of the following shaded region in xy-plane defined by $\{(x, y) : x \ge 0,\ 2y - x \ge 0,\ ax + y - 3 \le 0\}$ is equal to 3, then the value of a, lies in :

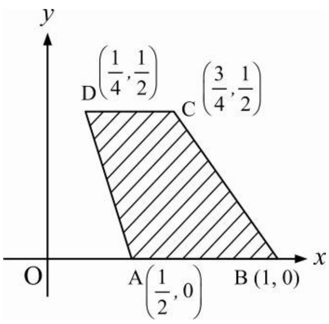
(A)



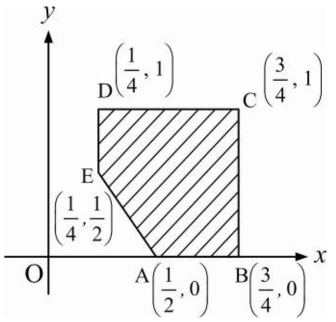
(B)



(C)



(D)



Section 2	Link Comprehension Type

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

Paragraph for Questions 31 - 32

A differentiable function satisfy the relation $ln(f(x+y)) = \ell n(f(x)) \cdot \ell n(f(y)) \ \forall\ x, y \in R,\ f(0) \neq 1,\ f'(0) = e$ and g be the inverse of f.

31. The value of $g''(e)$ is equal to :

(A) $\frac{-2}{e^2}$ **(B)** $\frac{-1}{2e^2}$ **(C)** $-2e^2$ **(D)** $\frac{2}{e^2}$

32. The number of solution(s) of the equation $\ell n[\ell n\, f(x)] = g^2[f(x)] - 3g[f(x)] - 5$ is(are) :

(A) 0 **(B)** 1 **(C)** 2 **(D)** Infinite

Paragraph for Questions 33 - 35

Let $f(x) = \begin{cases} 1-x & \text{if } 0 \le x \le 1 \\ 0 & \text{if } 1 < x \le 2 \\ (2-x)^2 & \text{if } 2 \le x \le 3 \end{cases}$ and $F(x) = \int_0^x f(t)\,dt$ then :

33. The function F(x) is :

(A) increasing in (0, 1) and decreasing in (2, 3)
(B) decreasing in (0, 1) and increasing in (2, 3)
(C) decreasing in $(0, 1) \cup (2, 3)$
(D) increasing in $(0, 1) \cup (2, 3)$

34. Range of F(x) is :

(A) $\left[0, \frac{1}{2}\right]$ **(B)** $\left[0, \frac{1}{3}\right]$ **(C)** $\left[0, \frac{5}{6}\right]$ **(D)** [0, 1]

35. Area enclosed by the curve $y = F(x)$ and x-axis and x varies from 0 to 3, is :

(A) 15/12 **(B)** 17/12 **(C)** 13/12 **(D)** 19/12

Paragraph for Questions 36 - 38

Consider the functions $f(x)$ and $g(x)$ such that $f(x) = \frac{x^3}{2} + 1 - x\int_0^x g(t)\,dt$ and $g(x) = x - \int_0^1 f(t)\,dt$. Both $f(x)$ and $g(x)$ are defined from $R \to R$.

36. Which one of the following holds good for $f(x)$?

(A) $f(x)$ is bounded
(B) $f(x)$ has exactly one maxima and one minima
(C) $f(x)$ has a maxima but no minima
(D) $f(x)$ has a minima but no maxima

37. Minimum distance between the functions $f(x)$ and $g(x)$ is :

(A) $\frac{4}{3\sqrt{2}}$ **(B)** $\frac{7}{6\sqrt{2}}$ **(C)** $\frac{7}{3\sqrt{2}}$ **(D)** $\frac{8}{3\sqrt{2}}$

38. The function $g(x)$:

(A) is injective but not surjective
(B) cuts the y-axis at $-\frac{3}{2}$
(C) cuts the y-axis at $\frac{3}{2}$
(D) is neither injective nor surjective

Paragraph for Questions 39 - 41

A curve $y = f(x)$ passing through origin and (2, 4). Through a variable point P(a, b) on the curve, lines are drawn parallel to coordinates axes. The ratio of area formed by the curve $y = f(x),\ x = 0,\ y = b$ to the area formed by the $y = f(x),\ y = 0,\ x = a$ is equal to 2 : 1.

39. Equation of line touching both the curves $y = f(x)$ and $y^2 = 8x$ is :

(A) $2x + y - 1 = 0$ **(B)** $2x + y + 1 = 0$ **(C)** $x + 2y + 1 = 0$ **(D)** $x + 2y - 1 = 0$

40. Pair of tangents are drawn from the point (3, 0) to $y = f(x)$. The area enclosed by these tangents and $y = f(x)$ is equal to :

(A) 9 **(B)** 18 **(C)** 15 **(D)** 27

41. AB is the chord of curve $y = f(x)$ passing through $\left(0, \frac{1}{4}\right)$. Locus of point of intersection of tangents at A and B is :

(A) $4y + 1 = 0$ **(B)** $4y - 1 = 0$ **(C)** $4x + 1 = 0$ **(D)** $4x - 1 = 0$

Paragraph for Questions 42 - 43

Let $f_n(x) = \sum_{n=1}^{n} \frac{sin^2 x}{cos^2\left(\frac{x}{2}\right) - cos^2\left(\frac{2n+1}{2}\right)x}$ and $g_n(x) = \prod_{n=1}^{n} f_n(x)\ \forall\ n = 1, 2, 3, \ldots\ldots$

42. Let $I_n = \int_0^{\pi} \frac{f_n(x)}{g_n(x)} dx$. If $\sum_{k=1}^{100} I_n = k\pi$, then the value of k is :

(A) 50 **(B)** 25 **(C)** 100 **(D)** 75

43. The value of $\left[\lim_{x \to 0} \int_0^x \frac{9dt}{x f_9(t) g_9(t)} \right]$ is :

(A) 50 **(B)** 10 **(C)** 100 **(D)** 25

Paragraph for Questions 44 - 47

Let $f(x)$ be a differentiable function such that $f(x+y) = e^x f(y) + e^y f(x)$ all x, y and $f'(0) = 1$.

44. $f(x)$ has :

(A) maximum **(B)** minimum
(C) both maximum and minimum **(D)** neither maximum nor minimum

45. The range of $f(x)$ is :

(A) R **(B)** $[0, \infty)$ **(C)** $\left(-\frac{1}{e}, 1\right)$ **(D)** $\left[-\frac{1}{e}, \infty\right)$

46. $\lim_{x \to -\infty} f(x)$ is :

(A) 0 **(B)** 1 **(C)** -1 **(D)** Non-existent

47. The area bounded by the curve $y = f(x)$ and the x-axis is :

(A) 1 **(B)** 1/2 **(C)** 2 **(D)** e

Section 3	**Multiple Correct Answer Type**

Each of the following Question has 4 choices A, B, C & D, out of which ONE OR MORE Choices may be Correct.

48. Let $f(x) = min(|e - x|, \pi - |x|)$. Then which of the following statement(s) is(are) correct ?

(A) $f(x)$ is many one but not even function

(B) Range of $f(x)$ is $\left(-\infty, \dfrac{\pi - e}{2}\right]$

(C) $f(x)$ is continuous and differentiable at all integral points

(D) $f(x)$ is continuous everywhere but non- differentiable at exactly two points

49. If $\lim\limits_{x \to 0} \dfrac{p\cos x + xe^{1/x}}{1 + \sin x + q\cos x . e^{1/x}} = 0$, then which of the following is(are) incorrect about p, q ?

(A) $p = 0, q \in R$ **(B)** $p = 4, q = 2$ **(C)** $p = 2, q \in R$ **(D)** $p = 0, q = 0$

50. Let $P(x)$ be a polynomial of n degree and $f(x) = \begin{cases} P\left(\dfrac{1}{x^3}\right)e^{-1/x^4} & , \quad x \neq 0 \\ 0 & , \quad x = 0 \end{cases}$, then :

(A) $f(x)$ is discontinuous at $x = 0$ **(B)** $f(x)$ is continuous at $x = 0$

(C) $f(x)$ is non-differentiable at $x = 0$ **(D)** $f'(0) = \lim\limits_{x \to 0} f(x)$

51. Let $f : (0, \infty) \to (-\infty, \infty)$ be defined as $f(x) = e^x + \ell nx$ and $g = f^{-1}$, then :

(A) $g''(e) = \dfrac{1 - e}{(1 + e)^3}$ **(B)** $g''(e) = \dfrac{e - 1}{(1 + e)^3}$ **(C)** $g'(e) = e + 1$ **(D)** $g'(e) = \dfrac{1}{e + 1}$

52. If $f(x) = \int \dfrac{x}{2} d\left(\dfrac{x^2 - 1}{x^2}\right)$ and $f(2) = \dfrac{1}{2}$ and $g_r(x) = \underbrace{f\left[f\left\{f\left(\ldots\ldots f(x)\ldots\ldots\right)\right\}\right]}_{r-\text{times}}$ i.e., $g_1(x) = f(x)$, $g_2(x) = f[f(x)]$ and so on then identify the correct statement(s).

(A) $\dfrac{d}{dx}\left(g_{(3n-2)}(x)\right) = 1$ whenever exists, $n \in N$

(B) $\dfrac{d}{dx}\left(g_{3n}(x)\right) = 1$ whenever exists, $n \in N$

(C) $\lim\limits_{x \to 1} \dfrac{\sum\limits_{r=1}^{100} \left(g_{3r}(x)\right)^r + x - 101}{x - 1} = 5050$

(D) Slope of the tangent to the graph of the function $y = g_{80}(x)$ at $x = \dfrac{1}{2}$ is 4

53. Let $f(x) = \int\limits_1^x \dfrac{3^t}{1 + t^2} dt,\ x > 0$ then which of the following is(are) false ?

(A) For $0 < \alpha < \beta$, $f(\alpha) < f(\beta)$ ~~**(B)** For~~ $0 < \alpha < \beta$, $f(\alpha) > f(\beta)$

~~**(C)** For all~~ $x > 0$, $f(x) + \dfrac{\pi}{4} < \tan^{-1} x$ ~~**(D)**~~ For all $x > 0$, $f(x) + \dfrac{\pi}{4} > \tan^{-1} x$

54. Let $f(x)=\int_0^x e^{-t^2}(t-5)(t^2-7t+12)dt$ for all $x\in(0,\infty)$, then :

(A) f has a local maximum at $x = 4$ and a local minimum at $x = 3$

(B) f is decreasing on $(3, 4)\cup(5, \infty)$ and increasing on $(0, 3)\cup(4, 5)$

(C) There exists atleast two $c_1, c_2 \in(0, \infty)$ such that $f''(c_1)=0$ and $f''(c_2)=0$

(D) There exists some $c\in(0, \infty)$ such that $f'''(c)=0$

55. Let $f(x)=\begin{cases} sin^{-1}(sin\,x) & , \quad x>0 \\ \dfrac{\pi}{2} & , \quad x=0 \\ cos^{-1}(cos\,x) & , \quad x<0 \end{cases}$, then :

(A) $f(x)$ has local maxima at $x = 0$ **(B)** $f(x)$ at continuous for all $x\in R$

(C) $f(x)$ has maximum value π **(D)** $f(x)$ has minimum value 0

56. If f is an odd continuous function in $[-1, 1]$ and differentiable in $(-1, 1)$, then which of the following statement(s) is(are) correct ?

(A) $f'(a) = f(1)$ for some $a\in(-1, 0)$

(B) $f'(b) = f(1)$ for some $b\in(0, 1)$

(C) $n(f(\alpha))^{n-1} f'(\alpha) = (f(1))^n$ for some $\alpha\in(-1, 0)$ and $\forall\ n\in N$

(D) $n(f(\beta))^{n-1} f'(\beta) = (f(1))^n$ for some $\beta\in(1, 0)$ and $\forall\ n\in N$

57. Let the function $f(x)$ be thrice differentiable and satisfies $f(f(x))=1-x$ for all $x\in[0,1]$. If $J=\int_0^1 f(x)dx$ and $f''\left(\dfrac{4}{5}\right)=0$, then which of the following is(are) true ?

(A) $f\left(\dfrac{1}{3}\right)+f\left(\dfrac{2}{3}\right)=1$ **(B)** $J=\dfrac{1}{2}$

(C) $f''(x)=0$ has at least one root in $x\in\left(\dfrac{1}{4}, \dfrac{3}{4}\right)$

(D) $f'''(x)=0$ has at least one root in $x\in\left(\dfrac{1}{2}, \dfrac{4}{5}\right)$

58. Consider $f(x)=e^x\,sec\,x-\sqrt{2}\,cos\,x+x,\ x\in\left[\dfrac{-\pi}{3}, \dfrac{\pi}{3}\right]$

(A) minimum value of $f(x)$ is $\left(\sqrt{2}e^{-\pi/4}-1-\dfrac{\pi}{4}\right)$

(B) minimum value of $f(x)$ is $\left(2e^{-\pi/3}-\dfrac{1}{\sqrt{2}}-\dfrac{\pi}{3}\right)$

(C) $f'\left(\dfrac{\pi}{3}\right)\ge f'(x)\ \forall\ x\in\left[\dfrac{-\pi}{3}, \dfrac{\pi}{3}\right]$ **(D)** $f'\left(\dfrac{\pi}{3}\right)\le f'(x)\ \forall\ x\in\left[\dfrac{-\pi}{3}, \dfrac{\pi}{3}\right]$

59. Let $f(x)=\dfrac{1-cos\{x\}}{\left(x^4+ax^3+bx^2+cx\right)^2}$. If $\ell=\lim\limits_{x\to1^+} f(x)$, $m=\lim\limits_{x\to2^+} f(x)$ and $n=\lim\limits_{x\to3^+} f(x)$, where ℓ, m and n are non-zero finite then : [where $\{x\}$ denotes fractional part of x]

(A) $\lim\limits_{x\to0^+} f(x)=\dfrac{1}{36}$ **(B)** $a+b+c=-1$

(C) $\ell+m+n=\dfrac{19}{72}$ **(D)** $\lim\limits_{x\to4^+} f(x)=0$

60. Let f be a differentiable function on R satisfying $f'(t)=e^t(cos^2 t - sin 2t)$ and $f(0)=1$, then which of the following is(are) correct ?

(A) f is bounded in $x \in (-\infty, 0)$

(B) Number of solution satisfying the equation $f(t)-e^t = 0$ in $[0, 2\pi]$ is 3

(C) The value of $\lim\limits_{t \to 0} (f(t))^{1/t} = 1$ **(D)** f is neither odd nor even

61. Let f be a differentiable function on R and satisfying the integral equation $\int_0^x f(t)dt + \int_0^x t.f(x-t)dt = -1+e^{-x}$ for all $x \in R$, then :

(A) $f(2)=e^{-2}$ **(B)** $f(0)+f'(0)=1$ **(C)** $f'(0)=2$ **(D)** $f'(0)=1$

62. Let $f : [0, \infty) \to R$ be a continuous strictly increasing function, such that $f^3(x) = \int_0^x t \cdot f^2(t)dt$ for every $x \geq 0$. Which of the following is(are) correct ?

(A) $f(6)$ is equal to 6 **(B)** $f(x)$ is surjective

(C) $\int_0^1 f(x)dx$ is equal to $1/18$ **(D)** Number of solutions of equations $f(x) = 6$ are two

63. A function $y=f(x)$ satisfying the differential equation $\frac{dy}{dx} \cdot \sin x - y\cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \to 0$ as $x \to \infty$ then the statement which is correct is :

(A) $\lim\limits_{x \to 0} f(x) = 1$ **(B)** $\int_0^{\pi/2} f(x)dx$ is less than $\frac{\pi}{2}$

(C) $\int_0^{\pi/2} f(x)dx$ is greater than unity **(D)** $f(x)$ is an odd function

64. Let $\frac{dy}{dx} + y = f(x)$ where y is a continuous function of x with $y(0)=1$ and $f(x) = \begin{cases} e^{-x} & \text{if } 0 \le x \le 2 \\ e^{-2} & \text{if } x > 2 \end{cases}$.
Which of the following hold(s) good?

(A) $y(1)=2e^{-1}$ **(B)** $y'(1)=-e^{-1}$ **(C)** $y(1)=-2e^{-3}$ **(D)** $y'(3)=-2e^{-3}$

65. If $y(x)$ satisfies the differential equation $\frac{dy}{dx} = sin 2x + 3y cot x$ and $y\left(\frac{\pi}{2}\right)=2$, then which of the following statement(s) is(are) correct ?

(A) $y\left(\frac{\pi}{6}\right)=0$ **(B)** $y'\left(\frac{\pi}{3}\right)=\frac{9-3\sqrt{2}}{2}$ **(C)** $y(x)$ increases in interval $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$

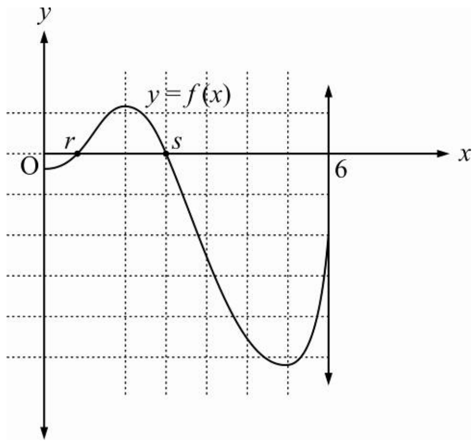
(D) The value of definite integral $\int_{-\pi/2}^{\pi/2} y(x)dx$ equals π

66. The graph of $y = f(x)$ is shown with roots r and $s (r < s)$. Area bounded by the graph of $f(x)$, x-axis, $x = 0$ and $x = 6$ over the intervals $[0, r]$ and $[r, s]$ and $[s, 6]$ are $\frac{2}{5}$, 2 and 12 respectively.

If $m = \int_s^0 f(x)dx,\ p = \int_r^6 f(x)dx,\ q = \left|\int_0^6 f(x)dx\right|, n = \int_0^6 |f(x)|dx$, then :

(A) $p < m$ **(B)** $m < 1$

(C) $q < 11$ **(D)** $n > 11$



Section 4	Match Matrix Type

Each of the following question contains statements given in two columns, which have to be matched. Statements in Column 1 are labelled as (A), (B), (C) and (D) whereas statements in Column 2 are labelled as p, q, r, s and t. More than one choice from Column 2 can be matched with Column 1.

67. **MATCH THE COLUMN :**

	Column 1		Column 2
(A)	Number of integral values of x satisfying $\left|x^2-9\right|+\left|x^2-4\right|=5$, is	**(p)**	0
(B)	Let $f(x)=\begin{cases}-5 & , \ -\infty < x \le -1 \\ x-4 & , \ -1 < x \le 6 \\ 2 & , \ 6 < x < \infty\end{cases}$, then $\lim\limits_{x \to n} \dfrac{[x]^2-13[x]+12}{(x-7)(x-6)}$ is equal to [Note : Where n is the number of integers in the range of $f(\lvert x \rvert)$, $[x]$ denotes greatest integer less than or equal to x]	**(q)**	2
(C)	If $\lim\limits_{x \to 0} \dfrac{a \sin x + bxe^m + 3x^2}{\sin x - 2x + \tan x}$ exists and has value equal to L, then the value of $\dfrac{b-L}{a}$, is equal to	**(r)**	3
(D)	Let $P_n = \prod\limits_{k=1}^{n}\left(1-\dfrac{1}{{}^{k+1}C_2}\right)$. If $\lim\limits_{n \to \infty} P_n$ can be expressed as lowest rational in the form $\dfrac{a}{b}$, then the value of $(b-a)$ is equal to	**(s)**	4
		(t)	Non-existent

68. Let $f(x)=(x-1)(x-2)(x-3)\ldots\ldots(x-n)$, $n \in N$ and $\displaystyle\int \frac{f(x)f''(x)-[f'(x)]^2}{f^2(x)}dx = g(x)+C$, where C is arbitrary constant. Match the following :

	Column 1		Column 2
(A)	If $f'(n)=5040$, then n is divisible by	**(p)**	4
(B)	If $g(x)$ is discontinuous at 9 points, $\forall\ x \in R$ then n is greater than	**(q)**	6
(C)	If $g(x)$ = 5 has 8 solutions, then n may be equal to	**(r)**	8
(D)	If the number of roots of equation $f'(x)=0$, be $(n-5)^2(n-1)$, $(n>1)$ then possible values of n is(are)	**(s)**	9

69. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	Let $f(x)=\begin{cases} x^{3/5} & \text{if } x\le 1 \\ -(x-2)^3 & \text{if } x>1 \end{cases}$ then the number of critical points on the graph of the function is	**(p)**	5
(B)	Number of real solution of the equation $log_2{}^2 x+(x-1)log_2 x=6-2x$, is	**(q)**	4
(C)	The number of values of c such that the straight line $3x+4y=c$ touches the curve $\frac{x^4}{2}=x+y$ is	**(r)**	3
(D)	If $f(x)=\int_x^{x^2}(t-1)dt,\ 1\le x\le 2$, then global maximum value of $f(x)$ is	**(s)**	2
		(t)	1

70. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	$\lim_{n\to\infty}\sum_{i=1}^{n}\frac{3}{n}.\,sin\left(2\pi+\frac{3\pi i}{n}\right)$ is equal to	**(p)**	$\frac{1}{e}$
(B)	$\int_0^{\infty}[x]\,e^{-x}dx$ equals (where [ ] denotes the greatest integer function)	**(q)**	$\frac{2}{e}$
(C)	If $\lim_{n\to\infty}\sum_{i=1}^{n}\left(\frac{\ell n 2i-\ell n\, n}{n}\right)=\ell n\, k$, then k equals	**(r)**	$\frac{1}{\pi}$
(D)	$\lim_{n\to\infty}\sum_{i=1}^{n}\frac{i}{n^2}sin\left(\frac{i^2\pi}{n^2}\right)$ is equal to	**(s)**	$\frac{2}{\pi}$
		(t)	$\frac{1}{e-1}$

71. MATCH THE FOLLOWING :

	Column 1		Column 2
(A)	Given that $f(0)=0,\ f'(0)=1,\ f'(2)=3$ and $f''(2)=5$. The value of the definite integral $\int_0^1 xf'''(2x)dx$ is equal to	**(p)**	2
(B)	The value of the definite integral $6\int_0^1\left(\frac{2^{log\,2^{x^4}}-3^{log\,27^{\left(x^2+1\right)^3}}}{7^{4\,log_{49}x}-x-1}\right)dx$ equals	**(q)**	6
(C)	If the value of the definite integral $\int_0^{2\pi}\left|x^2-\pi^2-sin^2 x\right|dx=k\pi^3$, then k equals	**(r)**	11

Section 5	Single Integer Answer Type

Each of the following question has an integer answer between 0 and 9.

72. A polynomial function P(x) of degree 5 with leading coefficient one, increases in the interval $(-\infty, 1)$ and $(3, \infty)$ and decreases in the interval (1, 3). Given that P(0) = 4 and $P'(2)=0$. If the value of $P'(6)$ = $abcd$, find the value of $a+b+c+d$.

73. Let $f(x)=\begin{cases}(x+2)^3 & , \quad -3<x\le -1 \\ x^{2/3} & , \quad -1<x<2\end{cases}$ and $g(x)=\int_{-3}^{x} f(t)\,dt,\ -3<x<2$. Find the number of extremum points of $g'(x)$.

74. The length of the shortest path that begins at the point (2, 5), touches the x-axis and then ends at a point on the circle $x^2+y^2+12x-20y+120=0$ is ab metre then a + b is ________.

75. Let $M=\left(p, \frac{4}{3-p}-1\right)$ be a variable point which moves in x-y plane. If $d=\sqrt{a}-\sqrt{b},\ a, b\in N$ is the least distance of the point M to the circle $(x-3)^2+(y+1)^2=1$, then find the value of $(a+b)$.

76. Let f be a twice differentiable function defined in $[-3, 3]$ such that $f(0)=-4,\ f'(3)=0,\ f'(-3)=12$ and $f''(x)\ge -2\ \forall \in[-3, 3]$. If $g(x)=\int_0^x f(t)\,dt$ then find maximum value of $g(x)$ is ab then $|a+b|$ is _______.

77. If $f(x)=a\,|\cos x|+b|\sin x|\,(a, b\in R)$ has a local minimum at $x=\frac{-\pi}{3}$ and satisfies $\int_{-\pi/2}^{\pi/2}(f(x))^2\,dx=2$. Find the values of a and b and hence find $\frac{b^2}{a^2}$.

78. Let $A(1,-1), B(4,-2)$ and C(9, 3) be the vertices of the triangle ABC. A parallelogram AFDE is drawn with vertices D, E and F on the line segments BC, CA and AB respectively. Find the maximum area of parallelogram AFDE.

79. Let $a_n\,(n\ge 1)$ be the value of x for which $\int_x^{2\pi} e^{-t^n}\,dt\,(x>0)$ is maximum. If $L=\lim\limits_{n\to\infty}\ell n(a_n)$ then find the value of e^{-L}.

80. Let f be a real function defined on R (the set of real numbers) such that $f'(x)=100(x-1)(x-2)^2(x-3)^3\ldots\ldots(x-100)^{100}$, for all $x\in R$. If g is a function defined on R such that $\int_a^x e^{f(t)}\,dt=\int_0^x g(x-t)\,dt+2x+3$, then find the sum of all the values of x for which g (x) has a local extremum is $abcd$ then $a+b+c+d$ is ________.

81. The value of $\lim\limits_{x\to 0}\left[\int_0^1 (by+a(1-y))^x\,dy\right]^{1/x}$ is $e^{-a}X$, then value of a is ________.

82. If $\dfrac{\int_0^1\left(1-\left(1-x^2\right)^{100}\right)^{201}.\,x\,dx}{\int_0^1\left(1-\left(1-x^2\right)^{100}\right)^{202}.\,x\,dx}=\dfrac{p}{q}$ where $p, q\in N$, then find the least value of $(p-q)$.

83. If x_1 and $x_2\left(x_1 < x_2\right)$ are two values of x satisfying the equation $\left|2\left(x^2+\frac{1}{x^2}\right)+\left|1-x^2\right|\right|=4\left(\frac{3}{2}-2^{x^2-3}-\frac{1}{2^{x^2+1}}\right)$ then find the value of $\int_{x_1+x_2}^{3x_2-x_1}\left\{\frac{x}{4}\right\}\left(1+\left[tan\left(\frac{\{x\}}{1+\{x\}}\right)\right]\right)dx$. [Note : $[y]$ and $\{y\}$ denote greatest integer fractional part functions respectively]

84. Given a curve C. Let the tangent line at P(x, y) on C is perpendicular to the line joining P and Q(1, 0). If the line $2x+3y-15=0$ is tangent to the curve C then the length of the tangent from the point (5, 0) to the curve C is $\sqrt{n}$ (where $n\in N$). Find the value of n.

85. A normal is drawn at a point P(x, y) on a curve. It meets the x-axis and the y-axis at A and B respectively such that (x-intercept)$^{-1}$ + (y-intercept)$^{-1}$ = 1, where O is origin, then find radius of the director circle of the curve passing through (3, 3).

86. Let $y=f\left(x\right)$ defined in [0, 2] satisfies the differential equation $y^3y''+1=0$ where $f\left(x\right)\geq 0\ \forall\ x\in D_f$ and $f'\left(1\right)=0,\ f\left(1\right)=1$ then find the maximum value of $f\,(x)$. [Note : D_f denotes the domain of the function, y'' denotes the 2nd derivative of y w.r.t. x]

87. Let $y=f\left(x\right)=\begin{cases}\sqrt{x+3} & , \quad -3\leq x<-2\\ 1+\sqrt{x+2} & , \quad -2\leq x<-1\\ 2+\sqrt{x+1} & , \quad -1\leq x\leq 0\end{cases}$. If $\left|y\right|=f\left(-\left|x\right|\right)$ be a curve and area enclosed between the curve and the circle $x^2+y^2=5$ equals $p+\pi q$, where p and q are integers then find the value of (p + q) is ab. Then a + b is __________.

88. The following figure shows the graph of a continuous function $y = f\,(x)$ on the interval [1, 3]. The points A, B, C have co-ordinates (1, 1), (3, 2), (2, 3) respectively and the lines L_1 and L_2 are parallel with L_1 being tangent to the curve at C. If the area under the graph of $y = f\,(x)$ from x = 1 to x = 3 is 4 square unit, then find the area (in same units) of shaded region.

89. Let $f\left(x\right)=cos^{-1}\left(\frac{2x}{1+x^2}\right),\ g\left(x\right)=cot^{-1}\left(\frac{2x}{x^2-1}\right)$ where $x\in\left(-1,1\right)$. If area bounded by the curves $y=f\left(x\right)+g\left(x\right)$ and $y=\pi x^2$ is A then find the value of [A]. [Note : [K] denotes greatest integer less than or equal to K]

90. If the area bounded by the curves $f\left(x\right)=\left[cos^{-1}\left|cos\,x\right|\right]^2,\ g\left(x\right)=\left[cos^{-1}\left|cos\,x\right|\right]$ and $\left|x\right|=\frac{\pi}{2}$ is $a\pi^3+b\pi^2+c$, then find the minimum value of $\left(\left|a\right|+\left|b\right|+\left|c\right|\right)$.

ANSWERS KEY

Advanced Mathematics Assignment-1A | 2019 | Calculus

Section 1	**Single Correct Answers Type**

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
B	B	C	B	B	B	B	C	C	D	D	D	B	A	C
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	C	C	C	B	C	A	A	D	B	A	C	C	D	B

Section 2	**Link Comprehension Type**

31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**
A	C	D	C	B	D	C	B	B
40	**41**	**42**	**43**	**44**	**45**	**46**	**47**	
B	A	A	C	B	D	A	A	

Section 3	**Multiple Correct Answer Type**

48	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**
CD	ABCD	BD	AD	BD	BCD	ACD	AC	ABD	ABCD
58	**59**	**60**	**61**	**62**	**63**	**64**	**65**	**66**	
AC	BCD	ABD	ABC	AC	ABC	ABD	AC	ABCD	

Section 4	**Matrix Match Type**

67	**68**	**69**
[A-s] [B-p] [C-r] [D-q]	[A-p, r] [B-p, q, r] [C-r] [D-p, q]	[A-r] [B-s] [C-t] [D-q]

70	**71**
[A-s] [B-t] [C-q] [D-r]	[A-p] [B-r] [C-p]

Section 5	**Single Integer Answer Type**

72	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**	**81**
3	2	4	7	4	3	5	2	7	1
82	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**	
1	2	3	4	1	6	3	4	1	